Praveen Rangavajhula

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Profile Summary

First-year PhD student in Computer Science at the University of Georgia with research interests in machine learning, predictive video modeling, longitudinal machine learning, optimization, learning dynamics, and interpretability. Experience includes V-JEPA-based latent future prediction, cognitive outcome modeling across longitudinal cohorts, automatic differentiation systems, privacy-preserving optimization, and federated learning.

Education

- **University of Georgia (UGA)** — *Athens, GA*
  Ph.D. in Computer Science — *Jan 2026 - Present*

- **University of Georgia (UGA)** — *Athens, GA*
  M.S. in Computer Science (GPA: 3.91) — *Aug 2023 - Dec 2025*

- **BITS Pilani** — *Pilani, India*
  B.E. in Mechanical Engineering — *Aug 2019 - July 2023*

Research Experience

- **Graduate Research Assistant** — *University of Georgia*
  Longitudinal Machine Learning for Cognitive Development using FFCWS and MIDUS — *Feb 2026 - Present*
  - **End-to-End ML Pipeline**: Owned the complete machine learning pipeline for predicting cognitive outcomes from longitudinal Future of Families and Child Wellbeing Study (FFCWS) data, including data ingestion, cross-wave alignment, preprocessing, train-only imputation, model implementation, evaluation, analysis, and visualization for 1,480 children across ages 3, 5, and 9.
  - **Model Development and Benchmarking**: Implemented and benchmarked nine model families: standard and feature-engineered linear regression, support vector regression, Random Forest, XGBoost, histogram gradient boosting, an attention network, TabICLv2, and a custom Neural CDE. Evaluated the models using five repeated child-level 80/20 train-test splits and 344 to 796 time-valid predictors.
  - **Empirical Findings**: Found that expanding from theory-driven features to hundreds of predictors increased TabICLv2's average held-out R^2 from 0.307 to 0.324, indicating limited incremental predictive value from the expanded feature space.
  - **Cross-Dataset Extension**: Extending the end-to-end pipeline to the Midlife in the United States (MIDUS) longitudinal study to examine whether the findings generalize to an independent cohort.

- **Master's Researcher** — *University of Georgia*
  Advised by Prof. John A. Miller — *Aug 2024 - Dec 2025*
  - **Custom Autograd Engine**: Designed and implemented a custom autograd engine in Scala, enabling automatic computation graph construction and backpropagation.
  - **Gradient Caching for Efficiency**: Integrated gradient caching mechanisms to store intermediate values, reducing redundant recomputation and improving training efficiency.
  - **Dynamic Computation Graph**: Implemented a dynamic graph structure that supports complex tensor operations, allowing for efficient forward and backward passes in deep learning models.
  - **Scalability and Modularity**: Developed reusable tensor and optimization modules, making the engine extensible for future integration with larger ML frameworks, including potential applications in large-scale model training.
  - **RNN and GRU Forecasting**: Manually derived backpropagation through time (BPTT) equations for RNNs and GRUs, analyzed gradient flow behavior, and applied gradient clipping to improve training stability.
  - **Implementation in Scala**: Implemented RNNs and GRUs from scratch using the derived equations, achieving a SMAPE reduction from 60% to 16% on forecasting tasks.
  - **Recurrent Architecture**: Designed modular `RecurrentBase` and `RNNCell` components for large-scale forecasting tasks.

- **Volunteer Researcher** — *University of Georgia*
  Advised by Profs. Fei Dou and Lakshmish Ramaswamy — *Jan 2025 - May 2025*
  - **Federated Learning for Carbon Footprint Tracking**: Implemented **Federated Learning** using **Flower** and **PyTorch** to track **carbon emissions of deep learning models (MobileNetV3, ResNet20)** across distributed nodes.
  - **Carbon-Aware Model Evaluation**: Analyzed trade-offs between **model accuracy, energy efficiency, and environmental impact** in federated deep learning settings.
  - **Edge AI Experimentation**: Conducted experiments on **Jetson devices** to evaluate federated learning models (MobileNetV3, ResNet20) under low-power, decentralized settings.

Skills

- **Programming Languages**: Python, Scala, Java, C, SQL
- **Machine Learning Libraries**: PyTorch, TensorFlow, Scikit-Learn, OpenCV
- **Data Science Tools**: NumPy, Pandas, Matplotlib, Seaborn, Jupyter, scikit-image
- **ML Techniques & Concepts**: Supervised Learning, Deep Learning, Representation Learning, Video Representation Learning, Latent-Space Prediction, Optimization, Backpropagation, Federated Learning, Differential Privacy, Autograd Systems

Projects

- **Predictive Video Representation Learning with V-JEPA 2** *Research Project*
  Video Future Selection and Latent Prediction *2026*
  - **V-JEPA Integration and Latent Prediction**: Built a JEPA-style video future-selection system using a frozen V-JEPA 2 encoder to extract normalized clip embeddings and a compact PyTorch latent predictor trained with a composite MSE and cosine loss to forecast next-step representations.
  - **End-to-End Data and Training Pipeline**: Developed MovingMNIST and real-video ingestion pipelines, standardized clips for the video encoder, generated K candidate futures per example, cached extracted embeddings, and implemented configurable AdamW training with validation-based best-state selection.
  - **Scoring, Baselines, and Evaluation**: Implemented frozen V-JEPA, learned latent-predictor, semantic, heuristic, and frozen-feature reranking methods; produced candidate rankings, confidence margins, and reproducible CSV/JSON benchmarks that revealed dataset-dependent trade-offs among scorers.
  - **Grounded Interpretability and Demonstration**: Created a deterministic explanation layer that converts structured scoring evidence into human-readable commentary, LLM-ready presenter prompts, and visualization-ready summaries for Jupyter demonstrations and research presentations.
- **Representation Learning Under Alternative Learning Rules** *University of Georgia*
  Independent Course Project in Representation Learning *Jan 2026 - May 2026*
  - **Learning Rule Implementations**: Independently implemented backpropagation, predictive coding, and Mono-Forward under matched multilayer perceptron architectures, conducting controlled experiments and validation-based model selection on MNIST and CIFAR-10.
  - **Representation Analysis**: Analyzed learned representations using layer-wise linear probes, t-SNE, and linear centered kernel alignment (CKA); Mono-Forward achieved 98.38% test accuracy on MNIST, 54.84% on CIFAR-10, and the strongest probe performance across both datasets.
  - **Representation Similarity**: Quantified learning-rule effects on representation geometry: backpropagation and Mono-Forward achieved a mean same-depth CKA of 0.749 on CIFAR-10, compared with 0.411 and 0.425 for pairings involving predictive coding.
  - **Memory Profiling**: Profiled peak CPU memory across matched MLPs with 2 to 10 hidden layers, observing lower measured peak memory for Mono-Forward as network depth increased.
- **Differentially Private Optimizer Research Project** *University of Georgia*
  Course Project in Privacy-Preserving Machine Learning (under Prof. Jaewoo Lee) *Aug 2024 - Dec 2024*
  - **DP-Lion: Private Optimizer Design**: Developed **DP-Lion**, a novel differentially private optimizer for deep learning, improving training stability and performance over DP-SGD and DP-Adam.
  - **Benchmarking on CIFAR-10**: Achieved **69% accuracy** on ResNet20 with privacy guarantees, outperforming most existing DP optimizers in empirical evaluations.
  - **Evaluation Against DeepMind Benchmarks**: Integrated techniques from DeepMind's private optimization research and compared DP-Lion against their reported baselines.
  - **Research Focus**: Used Opacus to privatize DP-Lion and baseline optimizers (DP-SGD, DP-Adam, DP-RMSprop), conducting empirical comparisons as part of a graduate privacy-preserving ML course project.
  - **Code**: GitHub Repository
- **Spotify Song Genre Classification** *University of Georgia*
  Course Project in Data Mining (under Prof. Fei Dou) *Aug 2024 - Dec 2024*
  - **Data Processing & Feature Engineering**: Processed **114,000 song samples**, extracting insightful **audio features** (e.g., **danceability-valence correlation**).
  - **Model Development**: Trained and compared **Decision Trees, Random Forests, SVM, kNN, and Neural Networks**, achieving **85.6% accuracy**.
  - **Experimentation & Metrics**: Designed and evaluated experiments using **precision-recall, ROC-AUC**, and **confusion matrices** to optimize classification.
  - **Code**: GitHub Repository
- **Diabetes Progression Prediction** *Kaggle Project*
  Data-Driven Modeling & Predictive Analytics *Sept 2024*
  - **Exploratory Data Analysis (EDA)**: Performed **data cleaning, feature engineering, and transformation (Box-Cox)** to improve regression model performance.

- **Predictive Modeling**: Built and evaluated a **Linear Regression model from scratch**, benchmarking against **Scikit-Learn** implementations.
- **Model Evaluation & Diagnostics**: Used **residual analysis, Q-Q plots, VIF checks**, and **homoscedasticity tests** to diagnose and improve model performance.
- **Regularization & Feature Selection**: Analyzed the impact of **Lasso, Ridge, and Elastic Net** regularization techniques to mitigate multicollinearity.
- **Code**: GitHub Repository

- **Spatio-Temporal Audio Classification** *University of Georgia*
  Course Project in Machine Learning in IoT (under Prof. Fei Dou) *Jan 2024 - May 2024*
  - **Model Adaptation**: Adapted a published **ConvLSTM-based architecture** to classify underwater acoustic signals on the **VTUAD dataset**.
  - **Learning Outcome**: Gained initial hands-on exposure to deep learning pipelines, model reproduction, and working with spatio-temporal data in ML.